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Milestones in Anatomy

- 1837 Hartig: Coined the term **Sieve tubes**
- 1839 Schleiden: Coined the term **Collenchyma**
- 1857 Hofmeister: Proposed **Apical cell theory**
- 1858 Nageli. C: Coined the term **Xylem and Phloem, Meristem and** supporter of **Apical cell theory**
- 1865 Mettenius: Coined the term **Sclerenchyma**
- 1868 Hanstein: Proposed **Histogen theory**
- 1885 Tschirch: Coined the term **Sclereids** Named Four types of **Sclereids** (Brachy, Macro, Osteo & Astro) in 1889
- 1914 Haberlandt: Coined the term xylem as **Hadrome** and Phloem as **Leptome and Classification of meristem.**
- 1924 Schmidt A: Proposed **Tunica – Corpus theory**
- 1926 Schűepp: **Mass, rib, & plate meristem**
- 1946 Bloch: Discovered the **Trichosclereids**
- 1952 Popham: Explained the organization of **Shoot apex of Angiosperms**
- 1955 Duchaigne: Discovered the **Annular collenchyma**
- 1961 Clowes: Proposed **Quiescent centre concept**
- 1963 Sanio: Coined the term **Tracheids**

The Tissues

A Tissue is a group of cells that are alike in origin, structure and function. The study of tissue is called Histology. A plant is made up of different types of tissues.

There are two principal groups:

1. Meristematic tissues
2. Permanent tissues

9.1 Meristematic Tissue

9.1.1 Characteristics and classification

The characters of meristematic tissues: (Gr. *Meristos*-**Divisible**)

The term meristem is coined by **C. Nageli** 1858.

- The meristematic cells are isodiametric and they may be, oval, spherical or polygonal in shape.
- They have generally dense cytoplasm with prominent nucleus.
- Generally the vacuoles in them are either small or absent.
- Their cell wall is thin, elastic and essentially made up of cellulose.
- These are most actively dividing cells.
- Meristematic cells are self-perpetuating.

Classification of Meristem

Meristem has been classified into several types on the basis of position, origin, function and division.

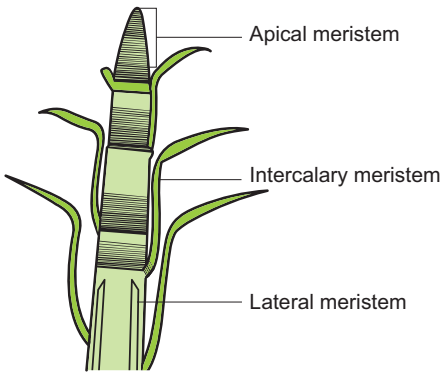


Figure 9.1: Different types of meristems on the basis of position in plant body